

Using AI to Identify High-Intent eCommerce Sessions Earlier



Identify intent
early



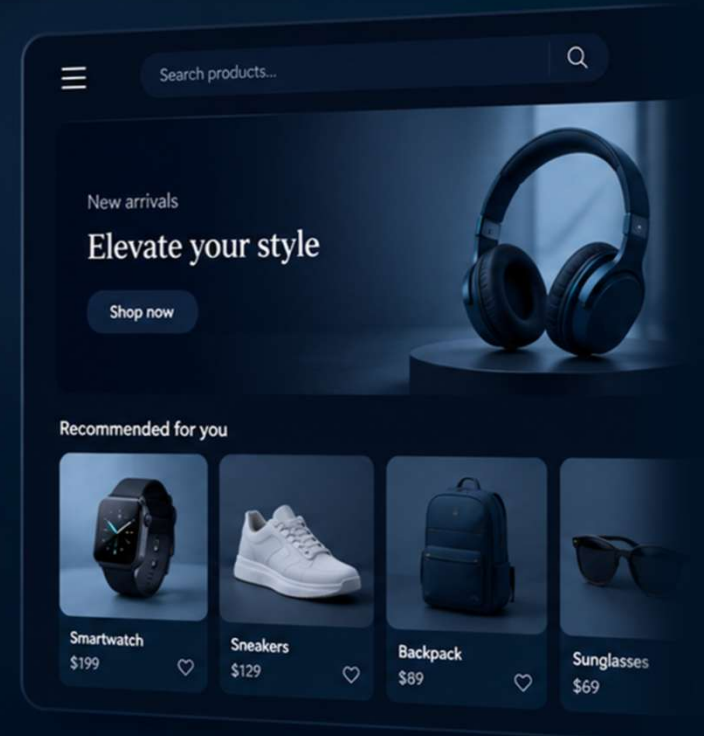
Personalize
in the moment



Increase
conversions

A Capstone Project by
Dominic L. Gacad

Using the Kaggle Dataset: eCommerce behavior data from multi category store (2020-Apr.csv)



Session Start
New Visitor



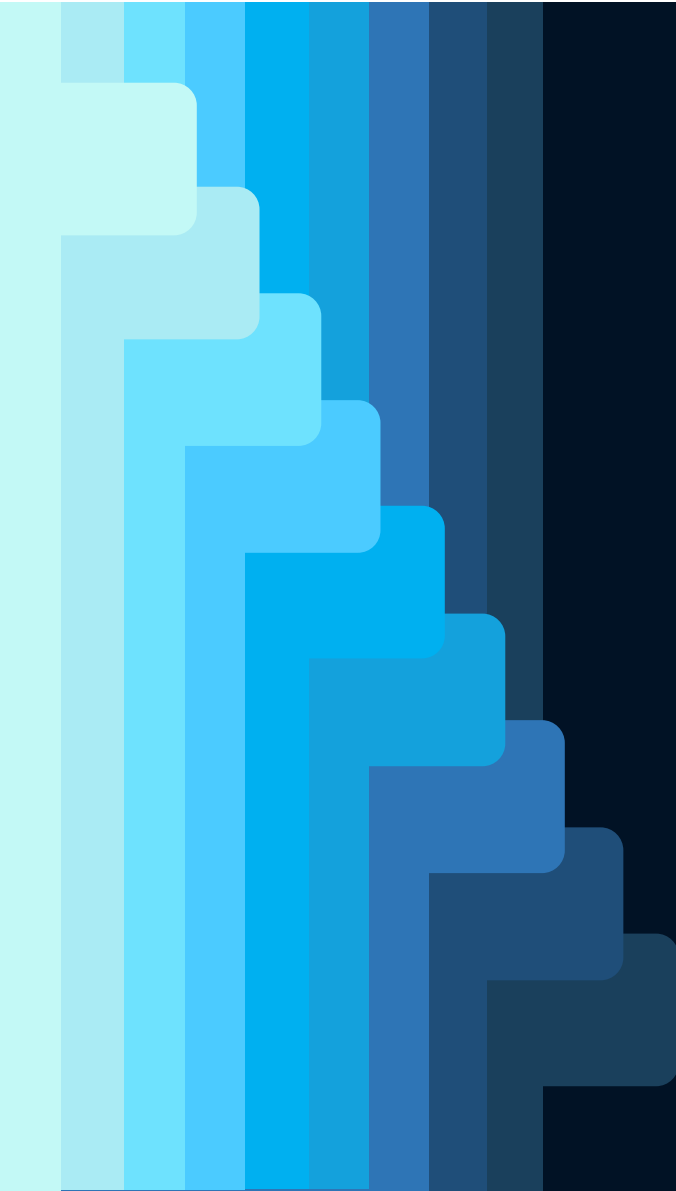
Browse
Explore



Engage
View Products



Signals Detected
AI Analysis



executive summary

built an AI model that can
identify likely future purchasers
after two early actions

61.38% precision
vs
13.41% baseline purchase rate

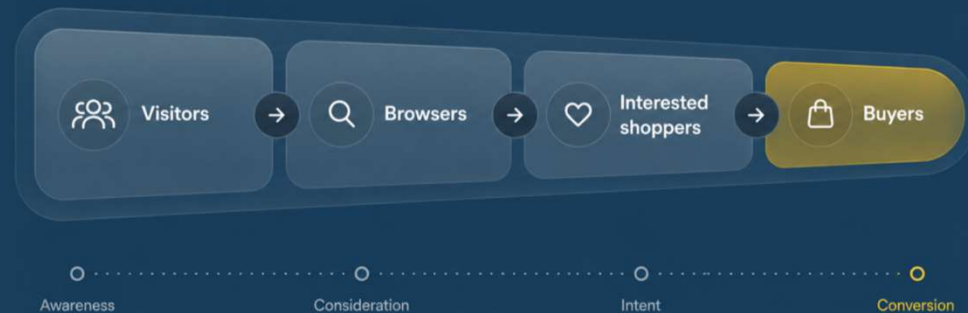
- *Predicts purchase intent early in the customer journey*
- *Helps prioritize marketing or remarketing efforts*

business problem

not all website visitors are equally likely to buy, and marketing resources should focus on the highest-intent sessions.

Simple Customer Funnel

From first visit to loyal customer



- *eCommerce traffic contains many low-intent sessions*
- *Generic targeting wastes marketing budget*
- *The goal is to identify high-intent sessions early*
- *Earlier detection allows faster engagement*

AI Purchase Intent Flow

Turning customer behavior into smarter engagement



- *observe the first two actions in a session*
- *look at what the customer viewed or added to cart*
- *consider timing, product type, brand, and price*
- *gives a purchase-likelihood score*

AI model
approach

the model uses only the first two customer actions to estimate purchase likelihood.



what data
was used

the model was trained on a large sample of real eCommerce behavior.



Dataset Overview

- Original dataset scanned: **66.6M events**
- Working sample: **982.6K events**
- Final modeling sessions: **92.8K**
- Final test set: **13,920 sessions**

- *No gender, race, age, income, or sensitive demographic fields were available or used.*

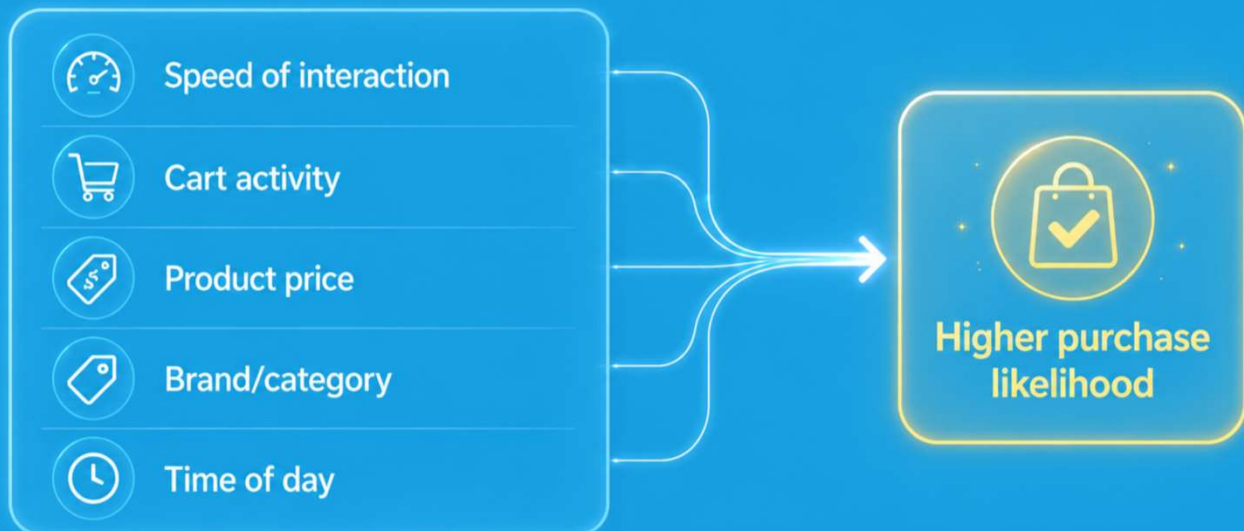
what the model learned

High-intent sessions show faster movement, stronger cart behavior, and meaningful product context.



Top Signals That Suggest Possible Purchase

Behavioral and contextual cues used by the AI model



model performance

the model identifies a smaller group of high-probability sessions with strong precision.

Model Performance Summary

AI Purchase Intent Modeling

	Test sessions	13,920
	Targeted sessions	1,489
	Correct future-purchase sessions identified	914
	Precision	61.38%
	Recall	48.96%
	F1-score	0.5447

From all test sessions to high-intent people and true future purchasers





Business Impact of the Purchase Intent Model

How model targeting improves marketing efficiency



Overall test
purchase rate:

13.41%



Model-targeted
precision:

61.38%



Approximate lift:

4.6x

business
value

the model can improve targeting efficiency by focusing on sessions with much higher purchase likelihood.

Useful for:



remarketing



limited-time offers



cart recovery



customer engagement
prioritization

Key considerations



- The model misses some future purchasers



- It favors immediate intent, especially early cart behavior



- It should not label unselected customers as low value



- It should not be used to deny offers or reduce service quality



- It needs monitoring over time

risks and responsible use the model should guide prioritization, not replace business judgment.

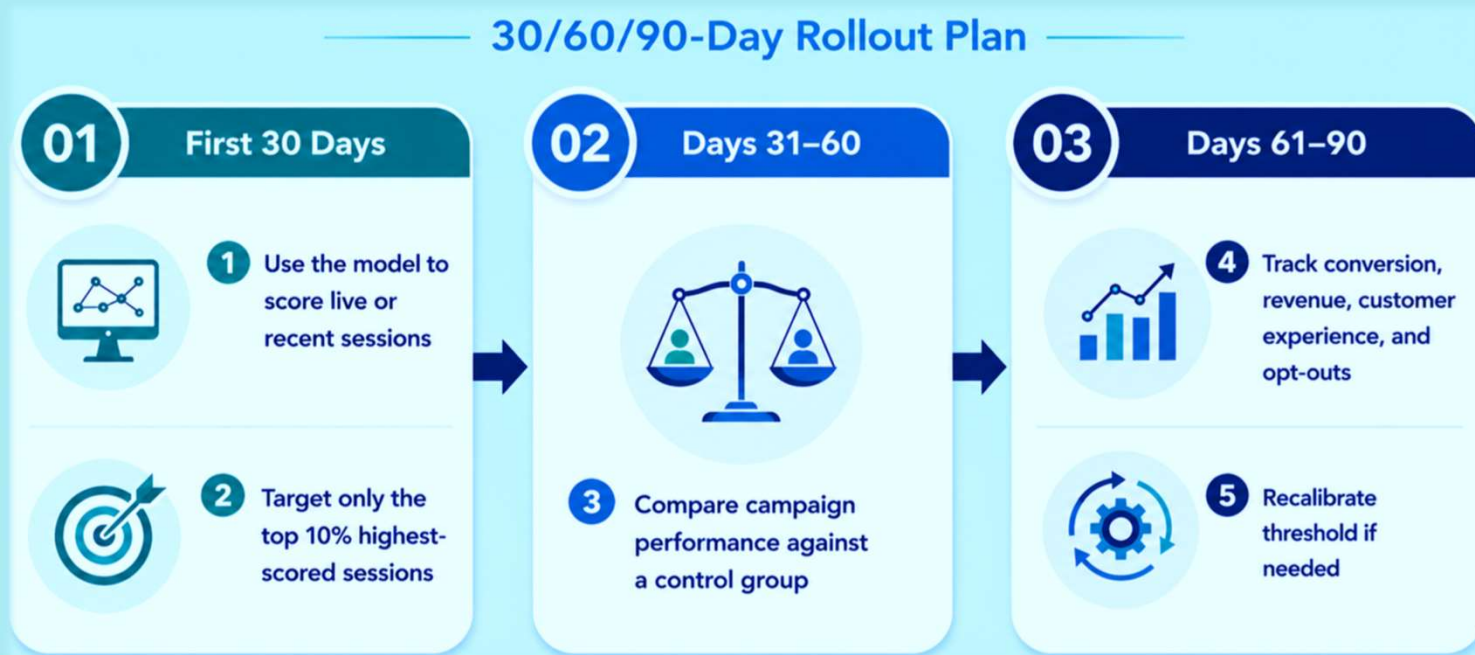
Risk-control table



Risk		Control	
	Missed buyers		Keep broad campaigns available
	Over-targeting		Use frequency caps
	Behavior drift		Monitor monthly
	Misuse		Require human oversight

recommended rollout

start with a controlled
pilot before production
deployment.



final recommendation

Proceed with a pilot using the model as a decision-support tool for targeted engagement.



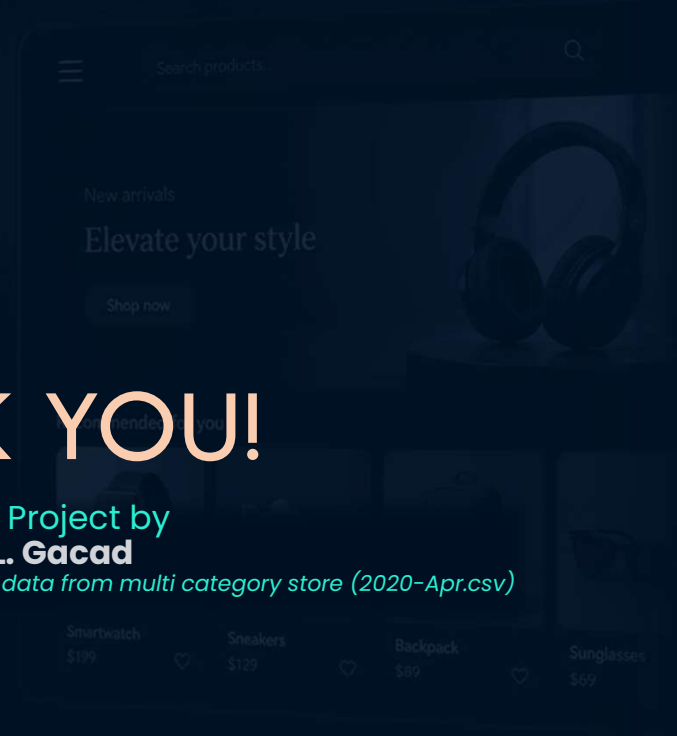
“The model is not a replacement for business strategy. It is a tool to help teams identify high-intent sessions earlier and act more efficiently.”

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THANK YOU!

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